
PROTOCOL

1. **Photograph the plates twice — blue-light transillumination and ambient light.** Save both; once you have picked, these are the only record.
2. **Tubes or a block.** Up to **4 cultures**: snap-cap tubes, each labelled with its clone ID. More than that: a **24-well block**, labels on the **airpore sheet** — the wells are identified by position.
3. Fill each tube or well with **4 mL 2YT + antibiotic** — the one on your labsheet. **It is not always carb.**
4. Pick **4 colonies from each plate**: touch a sterile toothpick to a single colony and drop it in. It stays there.
 - **Pick ones that are easy to pick and look healthy** — well separated, round and smooth. Big and well isolated is usually best. Contorted or smudgy means water seepage, or two colonies grown into one.
 - **Which colonies to choose is set by the experiment** — brightest, smallest, largest, red, or deliberately naive. **Your labsheet says which.**
 - **Into a block, fix the order before you start and write it down.** Wells go by position, so if your record and the block disagree the results are silently scrambled — nothing errors. Use your labsheet grid if it has one; **if not, draw one yourself** and keep it with the block. Tubes do not have this problem.
5. **Cover** — tubes: caps on loosely so they breathe; block: the airpore sheet — and grow in the **shaking incubator, 16–36 h**. Fridge by **36 h** or they overgrow.
6. **Parafilm the plates**, store them **upside-down in the fridge** — you may need them again.